

Phillips Curve

Intermediate Macroeconomics

Fall 2025, Sciences Po. Paris Campus du Havre

Lecture 6

Announcements

- Textbook: Chapter 8
- Quiz 2 will be open on October 9th 2025, and due on October 17th, 2025 (11:59PM).
- Mid-term exam will be open on Oct 17th and due 23rd (11:59 PM).

A brief history of the Phillips Curve:

- In 1958, A. W. Phillips found **a negative relation between inflation and unemployment**
 - UK data, 1861–1913
- In 1960, Paul Samuelson and Robert Solow
 - repeated the exercise for the US data, 1900–1960
 - labeled this relation the Phillips curve
- Since then, the Phillips Curve became central to macroeconomic thinking and policy
- In 1970s, the relationship broke down: periods of high inflation and high unemployment
- The relation reappeared as unemployment and *change* in the inflation rate
- It is widely used until today in its many mutations
 - Expectations, output gaps, acceleration etc.

The Original Phillips Curve

The UK data: 1861–1913

Describing a historical inverse relationship between rates of unemployment and corresponding rates of rises in wages

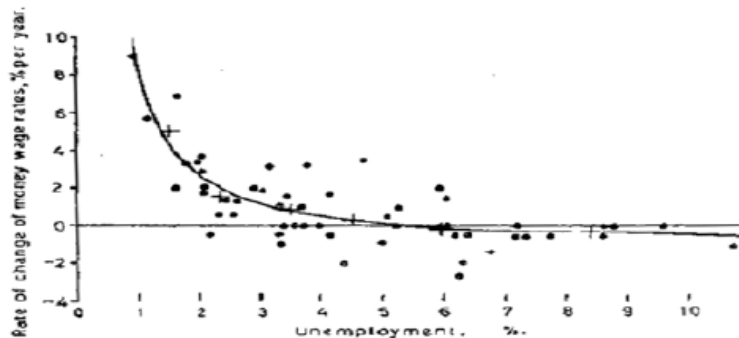
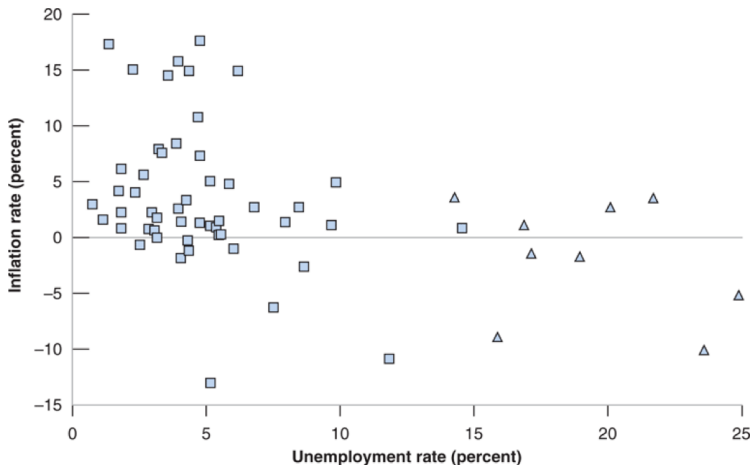


Fig.1. 1861 – 1913

Paul Samuelson and Robert Solow

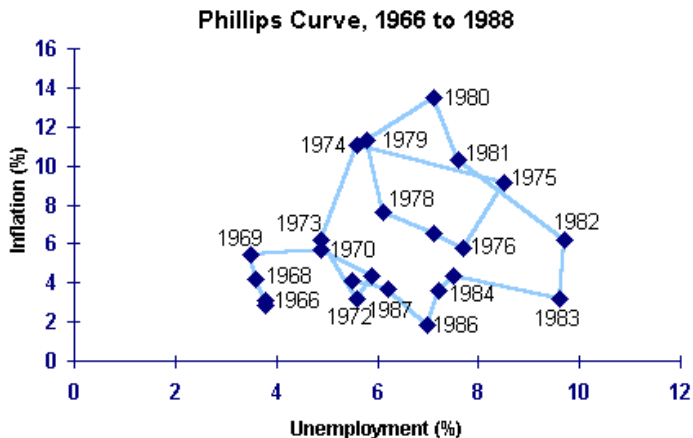
The US data: 1900–1960

A low unemployment rate was typically associated with a high inflation rate, and a high unemployment rate was typically associated with a low or negative inflation rate.

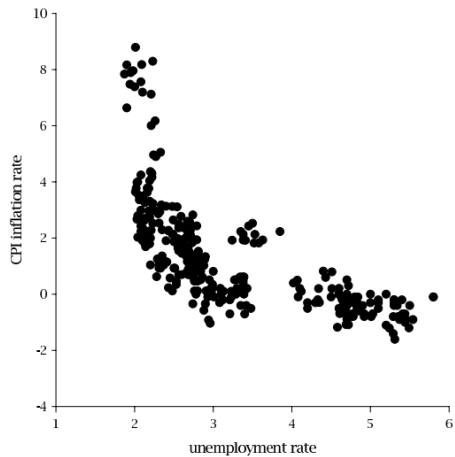


Breakdown of the Phillips Curve

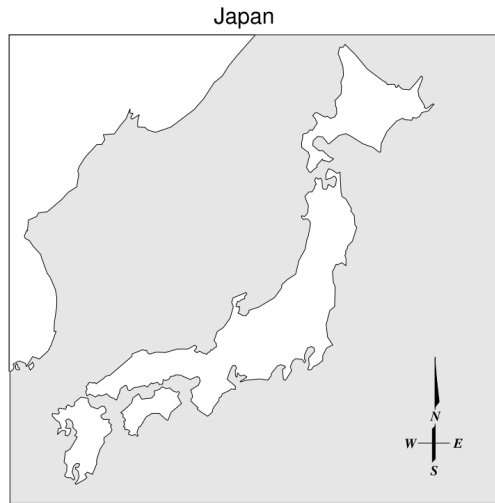
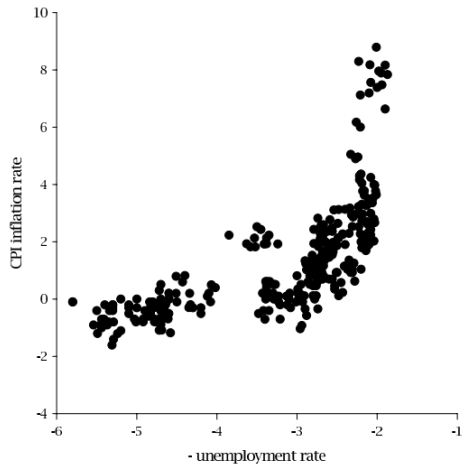
In the 1970s and early 1980s the short-run relationship between inflation and unemployment seemed to break down.



The Phillips Curve of Japan



Fun fact: The Phillips Curve of Japan



Today's lecture

1. Derive our Phillips curve (PC) from the Labor market equilibrium
2. Introduce various mutations of the PC
3. Link the PC with the natural rate of unemployment
4. Discuss policy implications (and many warnings)

Deriving the PC

Deriving the PC (full version in Appendix Chapter 8)

Start with the labor market (last lecture):

$$WS : W = P^e F(u, z)$$

$$PS : P = (1 + m)W$$

Substitute W in PS for WS equation:

$$P = P^e(1 + m)F(u, z)$$

Assume a specific form for F:

$$F(u, z) = 1 - \alpha u + z$$

The relation between the price level, the expected price level, and the unemployment rate is:

$$P = P^e(1 + m)(1 - \alpha u + z)$$

In terms of inflation rate π and the expected inflation rate π^e :

$$\pi = \pi^e + (m + z) - \alpha u$$

Deriving the PC

$$\pi_t = \pi_t^e + (m + z) - \alpha u_t$$

- An increase in inflation expectations π^e leads to an increase in π
- Given π^e , an increase in mark-ups (m), or an increase in other institutional and structural factors (z), leads to an increase in π
- Given π^e , a decrease in unemployment u leads to an increase in π .

The PC and its Mutations

The Early Version

$$\pi_t = \bar{\pi} + (m + z) - \alpha u_t$$

- We assume that $\pi_t^e = \bar{\pi}$
- Inflation varies from year to year around some value $\bar{\pi}$
- Negative relation between unemployment and inflation
- This Phillips curve relation was observed in the US in the 1960s
- The relation vanished in the 1970s because wage setters changed the way they formed inflation expectations

Inflation expectations

$$\pi_t^e = (1 - \theta)\bar{\pi} + \theta\pi_{t-1}$$

Expected inflation this year π_t^e depends on a combination of

1. A constant value $\bar{\pi}$
2. Inflation last year π_{t-1}

The combination of 1. and 2. is given by the weights θ and $(1 - \theta)$ where $0 < \theta < 1$ Extreme cases:

- if $\theta = 0$, $\pi_t^e = \bar{\pi}$ (as in the early versions)
- if $\theta = 1$, $\pi_t^e = \pi_{t-1}$ (expectations fully depend on past inflation)

Accelerationist PC

Let's assume that $\theta = 1$:

$$\pi_t^e = \pi_{t-1}$$

The Phillips curve is then:

$$\pi_t = \pi_{t-1} + (m + z) - \alpha u_t$$

Reshuffle the equation:

$$\underbrace{\pi_t - \pi_{t-1}}_{\Delta\pi} = (m + z) - \alpha u_t$$

Expectations-augmented Phillips curve, or the **accelerationist Phillips curve**

Based on the US data (1970–2014):

$$\Delta\pi = 3\% - 0.5u_t$$

Natural Rate of Unemployment

Natural Rate of Unemployment

is the unemployment rate at which the inflation rate remains constant.

Let's start with the PC:

$$\begin{aligned}\pi_t &= \pi_t^e + (m + z) - \alpha u_t \\ \pi_t - \pi_t^e &= (m + z) - \alpha u_t\end{aligned}$$

Assume that $\pi_t = \pi_t^e$:

$$0 = (m + z) - \alpha u_n$$

Solving for the natural rate u_n :

$$u_n = \frac{m + z}{\alpha}$$

The higher the markup (m) or other factor (z) that affect WS, the higher is the natural rate of unemployment

Natural Rate of Unemployment

Plug this back to the Phillips curve and we get:

$$\pi_t - \pi_t^e = -\alpha(u_t - u_n)$$

Let's take a simple case when $\pi_t^e = \pi_{t-1}$:

$$\underbrace{\pi_t - \pi_{t-1}}_{\Delta\pi} = -\alpha(u_t - u_n)$$

u_n is the natural rate of unemployment that keeps inflation rate constant:

Non-Accelerating Inflation Rate of Unemployment (NAIRU)

$$u_t = u_n \implies \Delta\pi = 0$$

Policy implications

PC and many unknowns (Book Chapter 8-4)

- Cross-country differences: labor-market rigidities
- Variation of natural rate over time ▶ The case of the US
- Wage indexation
 - Is a provision that automatically increases wages in line with inflation
 - Wage indexation increases the effect of unemployment on inflation.
- Deflation
 - When low inflation or deflation occurs, the Phillips curve relation breaks down.
 - One possible reason is the reluctance of workers to accept cuts in nominal wages.

Can policymakers misuse the PC?

$$\pi_t = \pi_t^e + (m + z) - \alpha u_t$$

Trade-off between inflation and unemployment

Can we engineer a lower unemployment rate if we are willing to tolerate higher inflation?

- Based on the PC: by increasing inflation, you can achieve a decrease in unemployment
- But(!) over time, people would learn to predict that you always increase inflation $\implies$ change in inflation expectations
- As a result, future increases in inflation would be fully anticipated ($\nearrow \pi^e$) and would not lead to decreases in unemployment
- This is one of the reasons why we have independent central banks

Extra

Variation of natural rate over time

Possible explanations for the decrease of the U.S. natural rate of unemployment from 7 to 8% in the 1980s to about 5% today:

- Increased globalization and stronger competition between U.S. and foreign firms may have lowered the markup.
- Increases in employment by temporary help agencies.
- The aging U.S. population and a decrease in the proportion of young workers and thus the unemployment rate.
- An increase in the incarceration rate.
- The increase in the number of workers on disability.

Derivation of PC: Appendix

$$P_t = P_t^e(1 + m)(1 - \alpha u_t + z)$$